

## Problem Set 5

### Antiderivatives

September 15, 2026

#### Problem 3

(j)

$$\begin{aligned} & \int \cos^3 x \, dx \\ &= \int \frac{\cos 3x}{4} + \frac{3 \cos x}{4} \, dx \\ &= \boxed{\frac{\sin 3x}{12} + \frac{3 \sin x}{4} + C} \end{aligned}$$

(k)

$$\begin{aligned} & \int \frac{(x^3 + 8)(x - 1)}{x^2 - 2x + 4} \, dx \\ &= \int (x + 2)(x - 1) \, dx \\ &= \int x^2 + x - 2 \, dx \\ &= \boxed{\frac{x^3}{3} + \frac{x^2}{2} - 2x + C} \end{aligned}$$

(l)

$$\begin{aligned} & \int \frac{1}{\sqrt{x+1} - \sqrt{x-1}} \, dx \\ &= \int \frac{\sqrt{x+1} + \sqrt{x-1}}{2} \, dx \\ &= \frac{1}{2} \cdot \frac{2}{3} \cdot ((x+1)^{\frac{3}{2}} + (x-1)^{\frac{3}{2}}) + C \\ &= \boxed{\frac{1}{3}(x+1)^{\frac{3}{2}} + \frac{1}{3}(x-1)^{\frac{3}{2}} + C} \end{aligned}$$

(m)

$$\int \sin^4 x \, dx$$

$$\begin{aligned}
&= \int \left( \frac{1 - \cos 2x}{2} \right)^2 dx \\
&= \frac{1}{4} \int \cos^2(2x) - 2 \cos 2x + 1 dx \\
&= \frac{1}{4} \int \frac{\cos 4x + 1}{2} - 2 \cos 2x + 1 dx \\
&= \int \frac{3}{8} - \frac{\cos 2x}{2} + \frac{\cos 4x}{8} dx \\
&= \boxed{\frac{3}{8}x - \frac{\sin 2x}{4} + \frac{\sin 4x}{32} + C}
\end{aligned}$$

(n)

$$\begin{aligned}
&\int \sin x \cos x \cos 2x \cos 4x \cos 8x dx \\
&= \int \frac{\sin 16x}{16} dx \\
&= \boxed{-\frac{\cos 16x}{256} + C}
\end{aligned}$$

**Problem 4**

(a)

$$\begin{aligned}
f'(x) &= 6\sqrt{x} + 5x^{\frac{3}{2}} \\
f(x) &= 4x^{\frac{3}{2}} + 2x^{\frac{5}{2}} + C \\
f(1) &= 10 \implies c = 4 \\
&\boxed{f(x) = 4x^{\frac{3}{2}} + 2x^{\frac{5}{2}} + 4}
\end{aligned}$$

(b)

$$\begin{aligned}
f''(\theta) &= \sin \theta + \cos \theta \\
f'(\theta) &= \sin \theta - \cos \theta + C \\
f'(0) &= 1 \implies C = 2 \\
f(\theta) &= -\cos \theta - \sin \theta + 2\theta + C \\
f(0) &= 2 \implies C = 3 \\
&\boxed{f(\theta) = -\cos \theta - \sin \theta + 2\theta + 3}
\end{aligned}$$

Two conditions are needed, as we are given the second derivative. Similarly, three are needed for the third derivative.

(c)

$$\begin{aligned}
 f'(x) &= \frac{3}{x^2} \\
 f(x) &= -\frac{3}{x} + C \\
 f(1) = 4 &\implies C = 7 \\
 \boxed{f(x) &= -\frac{3}{x} + 7}
 \end{aligned}$$

The answer is valid on  $(0, \infty)$ . For  $x < 0$ , as no initial condition is provided on this interval, no further progress can be made.

(d)

$$\begin{aligned}
 f'(x) &= 1 - \frac{1}{x^2} \\
 f(x) &= x + \frac{1}{x} + C \\
 f(1) = 0 &\implies C = -2 \\
 \boxed{f(x) &= x + \frac{1}{x} - 2}
 \end{aligned}$$

The three graphs have the same shape and slope, but they are vertical translations of each other. The theorem that guarantees this is theorem 3 (families of antiderivatives).

### Problem 5

(a)

$$\begin{aligned}
 v(t) &= 0.6t - 0.06t^2 \\
 v(5) &= 1.5m/s \\
 s(t) &= 0.3t^2 - 0.02t^3 \\
 s(5) &= 5m
 \end{aligned}$$

The elevator is speeding up on  $t \in [0, 5]$ .

(b)

$$\begin{aligned}
 v(t) &= -8t + 24 \\
 s(t) &= -4t^2 + 24t \\
 v(t) = 0 &\implies t = 3s \\
 s(3) &= 36m \\
 d &= \frac{v_t^2 - v_i^2}{2a} = \frac{v_o^2}{2k}
 \end{aligned}$$

(c)

$$\begin{aligned}
 v(t) &= -9.8t + 14 \\
 s(t) &= -4.9t^2 + 14t + 2 \\
 v(t) = 0 &\implies t = \frac{10}{7} \\
 s\left(\frac{10}{7}\right) &= 12m \\
 s(t) = 0 &\implies t = 2.99
 \end{aligned}$$

(d)

$$\begin{aligned}
 V(t) &= \int r(t) dt = -t^2 + 40t + C \\
 t(0) &= 150 \implies C = 150 \\
 V(t) &= -t^2 + 40t + 150 \\
 V(20) - V(0) &= 400 = A
 \end{aligned}$$

The net change equals the area under the curve.

## Problem 6

(a)

$$\begin{aligned}
 \frac{d}{dx}(\sin^2 x + C) &= 2 \sin x \cos x \\
 \frac{d}{dx}(-\cos^2 x + C) &= 2 \sin x \cos x \\
 \frac{d}{dx}\left(-\frac{1}{2} \cos 2x + C\right) &= 2 \sin x \cos x \\
 &-1, -\frac{1}{2}, \frac{1}{2}
 \end{aligned}$$

It is the theorem of equal derivatives.

(b)

$$\begin{aligned}
 \ln(x^2 + 1) + C &= \ln 3 + \ln(x^2 + 1) + C = \ln(3x^2 + 3) + C \\
 \ln 3 &\neq 0
 \end{aligned}$$

As  $C$  may represent any real value, the first  $C$  and second  $C$  simply have a difference of  $\ln 3$ , which absorbs the difference.

(c)

- (i) True, the difference of two constants is a constant.
- (ii) True, the constant  $C$  may vary.
- (iii) False, it obeys the product rule.
- (iv) False, missing constant  $C$ .

(d) The answers agree. The chain rule method for the case here works, as it only creates a constant term 2, while for  $(x^2 + 3)^5$  a linear term is added.

### Problem 7

(a)

$$\begin{aligned} F'(0^+) &= \frac{\frac{1}{2}h^2}{h} = 0 \\ F'(0^-) &= -\frac{\frac{1}{2}h^2}{h} = 0 \\ F'(0^+) &= F'(0^-) = 0 = f(0) \\ F'(x) &= |x| = f(x) \end{aligned}$$

(b)

$$\begin{aligned} x > 0 &\implies \frac{d}{dx} \ln x = \frac{1}{x} \\ x < 0 &\implies \frac{d}{dx} \ln -x = -\frac{1}{x} \\ \frac{d}{dx} \ln |x| &= \frac{1}{x} \\ G(x) &= \begin{cases} \ln x + 1, & x > 0 \\ \ln(-x), & x < 0 \end{cases} \\ F(x) &= \begin{cases} \ln x + C_1, & x > 0 \\ \ln(-x) + C_2, & x < 0 \end{cases} \end{aligned}$$

Theorem 2 fails as  $F(x)$  is not differentiable on the entire interval.

(c)

$$\begin{aligned} x \in (-1, 0] &\implies f(x) = 0 \implies F(x) = c \\ x \in (0, 1) &\implies f(x) = 1 \implies F(x) = x + k \\ \lim_{x \rightarrow 0^-} c &= \lim_{x \rightarrow 0^+} \implies c = k \\ F'(0^+) &= \frac{h}{h} = 1 \\ F'(0^-) &= \frac{0}{h} = 0 \\ F'(0^+) &\neq F'(0^-) \\ F'(x) &\neq f(x) \end{aligned}$$

(d) A function with a jump discontinuity skips values, and thus failing to meet the requirement of taking every value on an interval as a derivative. A step function is integrable, yet it does not have a derivative. Thus, having an anti-derivative is the stronger condition.